Bayesian Profile Regression for Multimorbidity Clustering at Population Scale

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Acknowledgements

Reference: `https://arxiv.org/abs/2602.24038`

HDRUK
Health Data Research UK

Project aims

- Find clusters of MLTC in EHRs held in SAIL
- Write a BPR model that can be fit using SVI
- Validate model performace using simulation studies

Bayesian Profile Regression

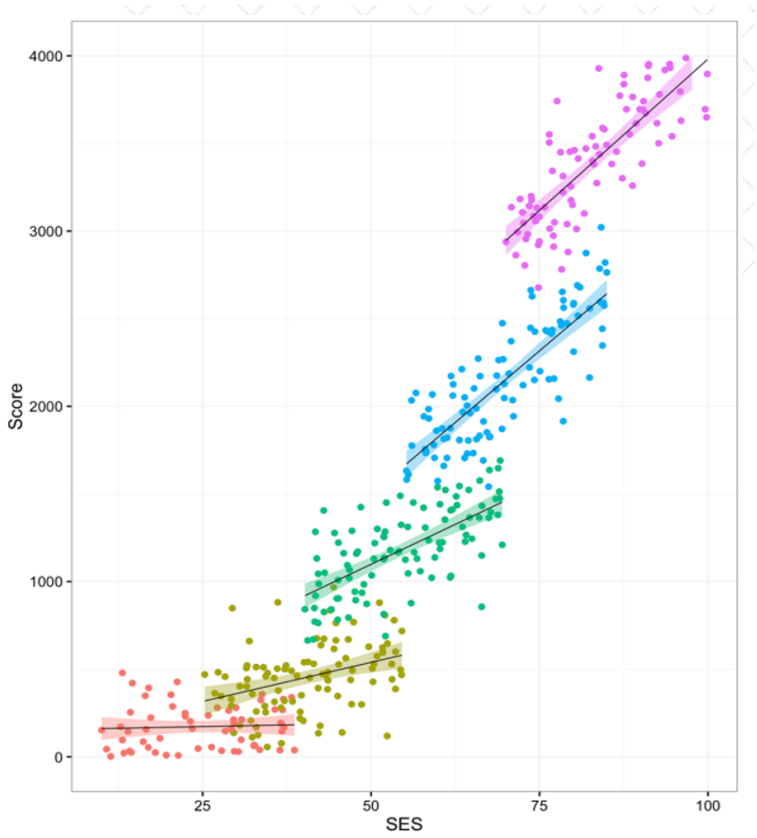
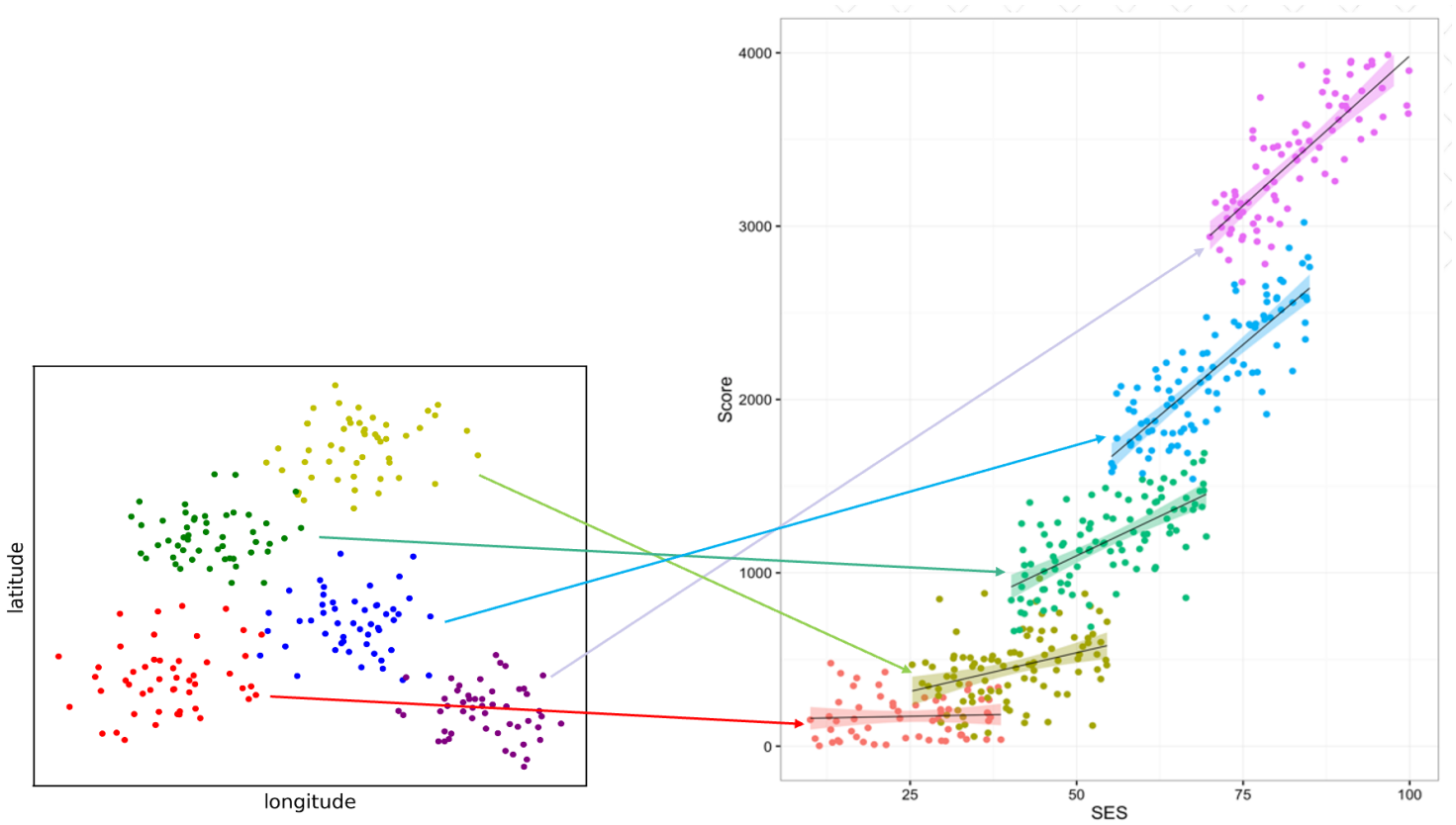


Image taken from https://www.r-bloggers.com/2016/10/multilevel-modeling-of-educational-data-using-r-part-1/ accessed 17th April 2026

Bayesian Profile Regression



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Aside - Dirichlet Process Mixture models

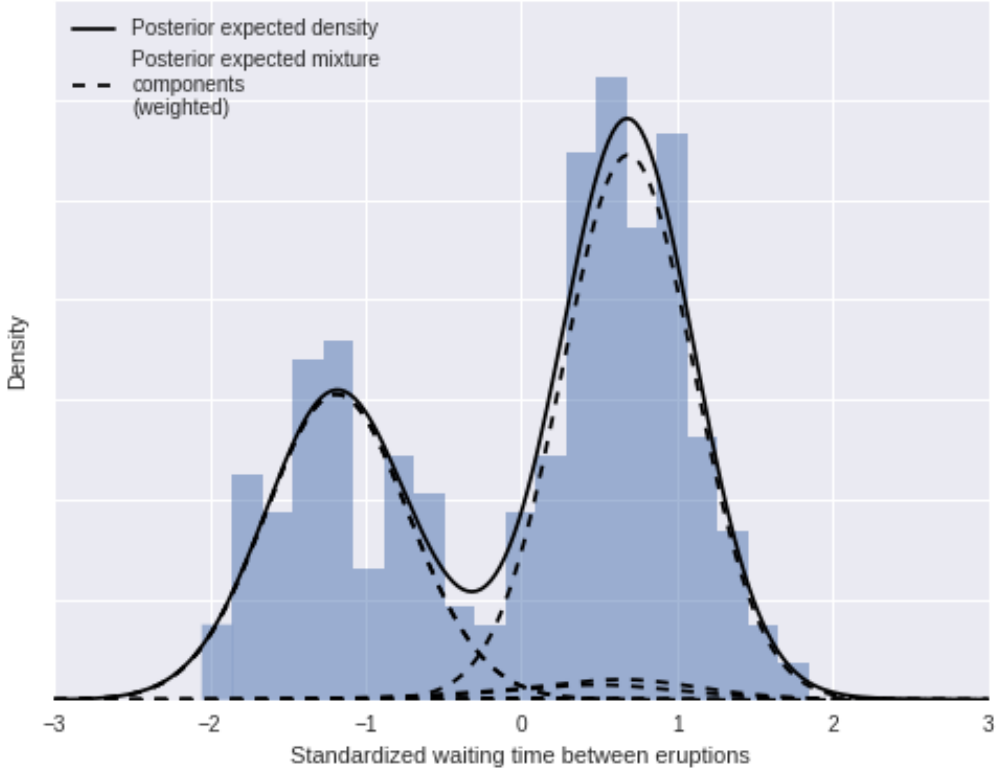


Image taken from https://austinrochford.com/posts/2016-02-25-density-estimation-dpm.html accessed 17th April 2026

Bayesian Profile Regression

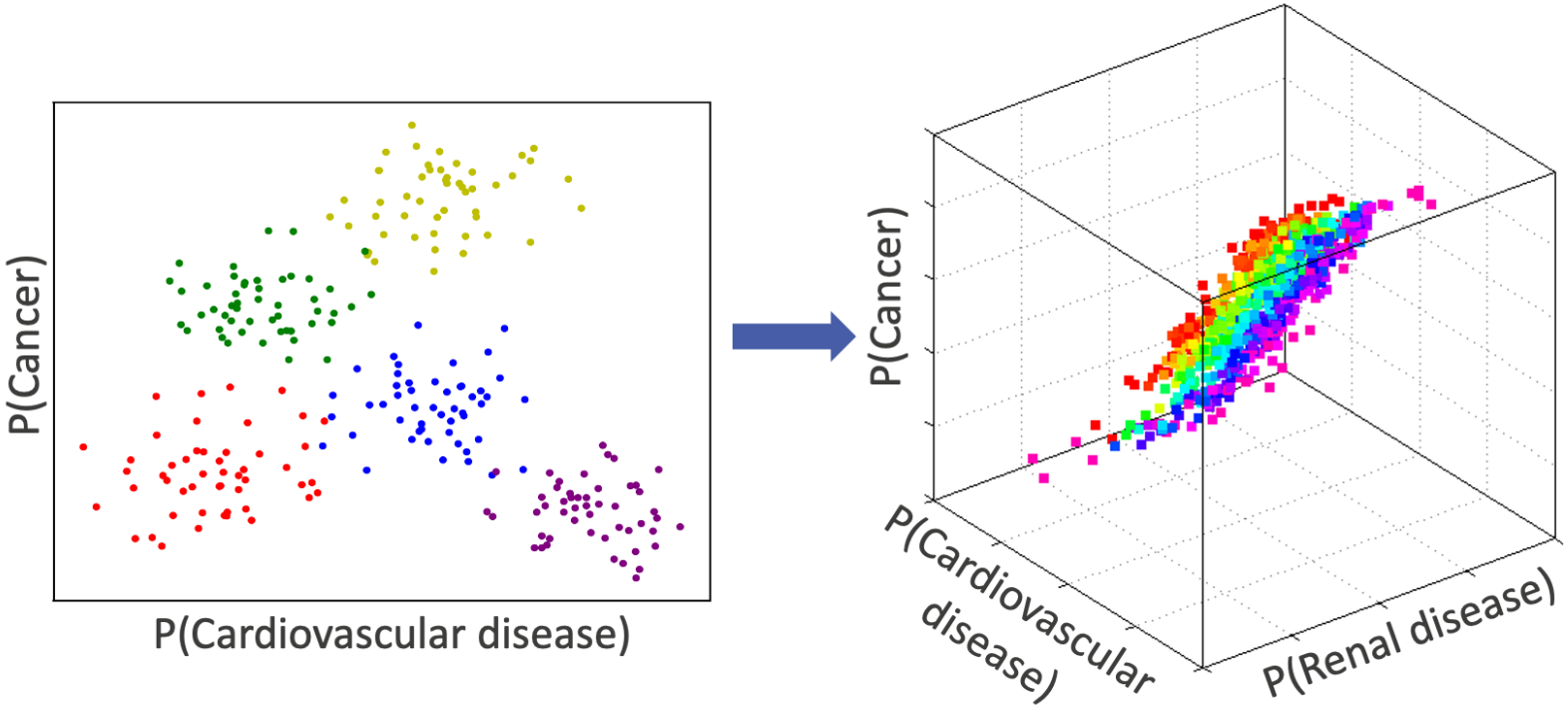


Image courtesy of Rhiannon Owen

References

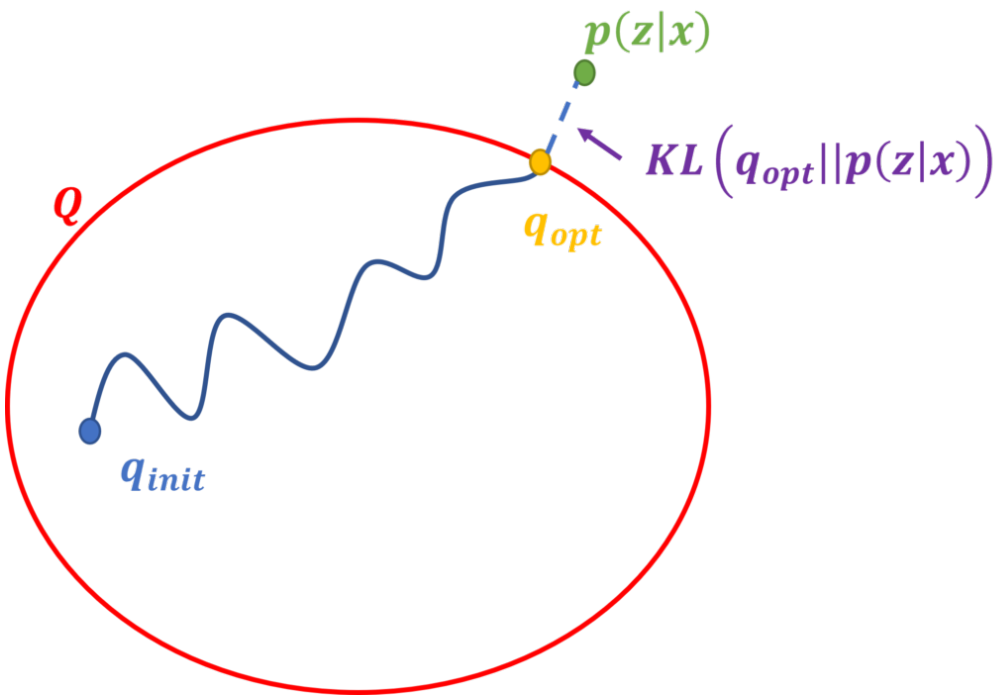
- Molitor J et al. Bayesian profile regression with an application to the National Survey of Children's Health. Biostatistics. 2010 Jul 1;11(3):484-98.
- Liverani S et al. PReMiuM: An R package for profile regression mixture models using Dirichlet processes. Journal of statistical software. 2015 Mar 20;64:1-30.

Standard fitting methods

- In a Bayesian framework we calculate the full posterior distribution rather than just point estimates
- Normally done with MCMC sampling.
- MCMC is computationally quite expensive.

Stochastic Variational Inference

Stochastic Variational Inference

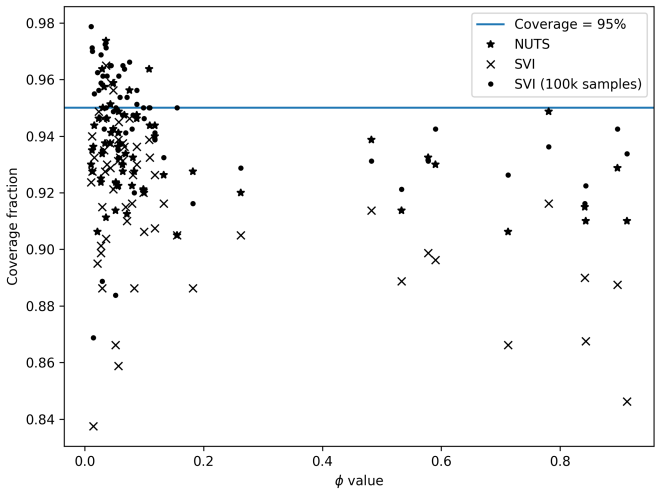


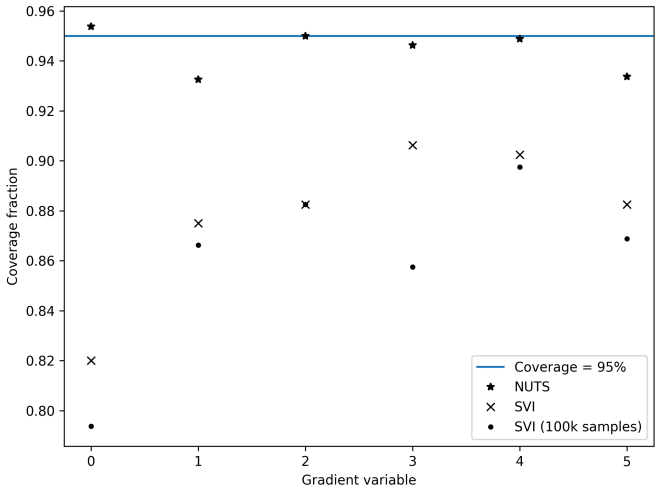
from https://meerkatstatistics.com/courses/variational-inference-in-r/ accessed 17th April 2026

Advantages & Disadvantages

No-U-Turn Sampler	Stochastic Variational Inference
✔ Convergence guarantees	✗ No convergence guarantees
✔ Convergence diagnostics	- Convergence diagnostics
✗✗ Computationally expensive	✔✔ Computationally efficient

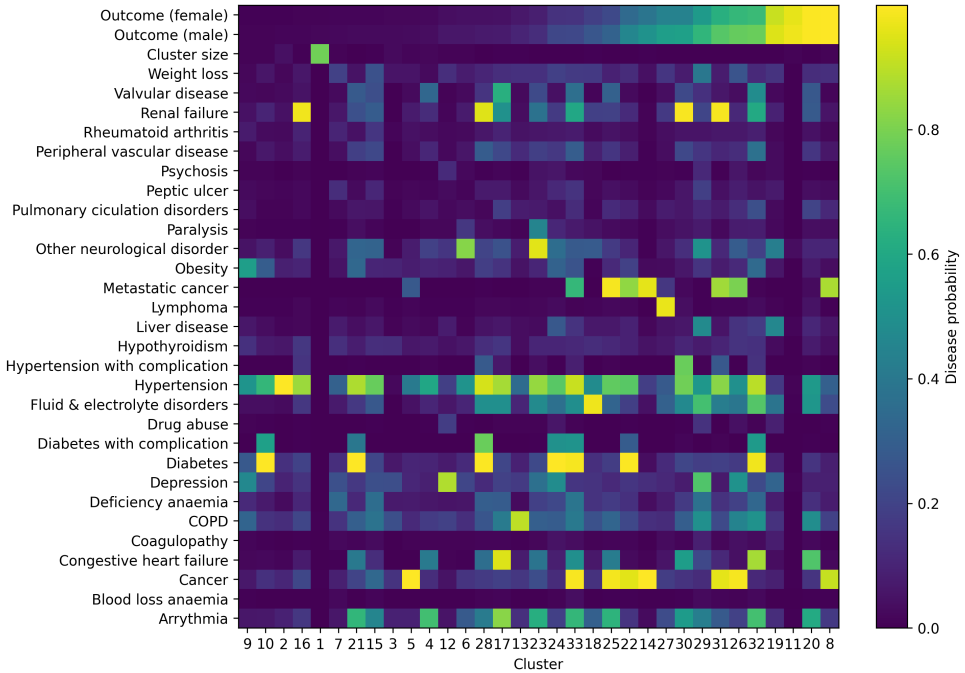
Results - Simulation study





Results - SAIL analysis

Variable	N(%) / Mean(SD)
Population N	1296463
Male Sex N(%)	631453 (48.70%)
Age Mean (SD) (years)	42.51 (15.95)
Died N(%)	160709 (12.40%)
Total Follow up (years)	20949059



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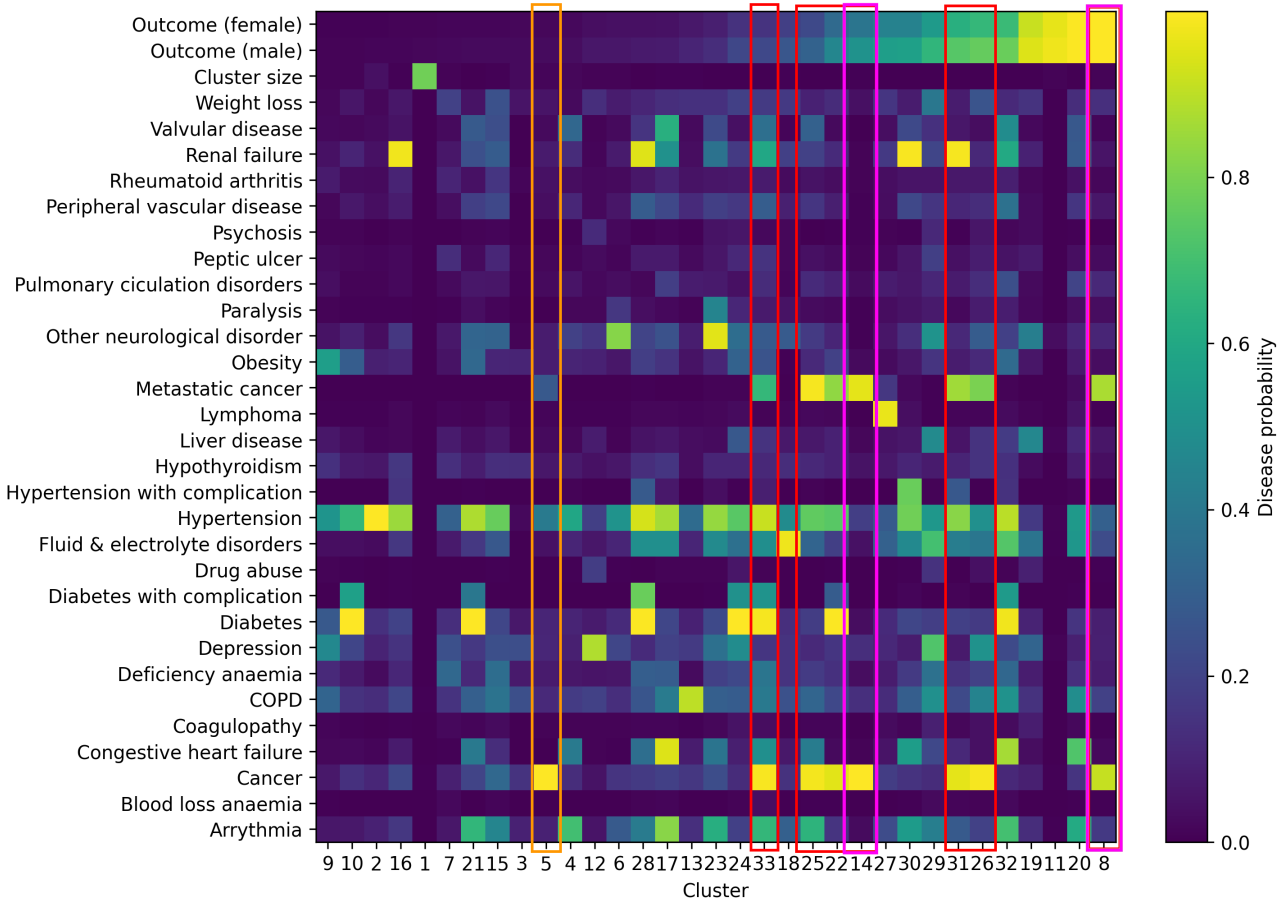

Summary & Future work

- BPR allows for granular clustering, conditioning on covariates.
- SVI allows Bayesian model fitting even at population-scale
- Future work
  - Time to event response model
  - More exotic mixture sector structures

Thank you

Questions?

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See `https://github.com/jim-rafferty/HDR_midlands_data_grand_rounds_20290421` for slides and notes

References:

JR et al. 2026 Bayesian Profile Regression using Variational Inference to Identify Clusters of Multiple Long-Term Conditions Conditioning on Mortality in Population-Scale Data `https://arxiv.org/abs/2602.24038`. Submitted to *Biostatistics*

J Molitor et al. 2010 Bayesian profile regression with an application to the National Survey of Children's Health. *Biostatistics*

S Liverani et al. 2015 PReMiuM: An R package for profile regression mixture models using Dirichlet processes. *Journal of statistical software*

J Lyons et al. 2021 Protocol for the development of the Wales Multimorbidity e-Cohort (WMC): data sources and methods to construct a population-based research platform to investigate multimorbidity *BMJ Open*

BPR - Model likelihood

$$p = p_M . p_R \tag{1}$$

$$p_M \left(x_{ij} | z_i = k\right)_{ij} \sim \text{Bernoulli}\left(\phi_j\right) \tag{2}$$

$$p_R(y_i | z_i = k, w_i)_i \sim \text{Bernoulli}\left(Y_i\right) \tag{3}$$

$$\text{logit}\left(Y_i\right) = \beta_{0k} + \sum_a \beta_a w_{ia} \tag{4}$$

Note: Dirichlet process omitted